

elimination of left recursion from the grammar.

Q. 3 Solve Any One of the following.

- A) Explain the importance of regular expression and automaton in the process of compilation. Convert the regular expression for valid variable in C programming language into minimized DFA using Firstpos, Lastpos and Followpos functions. (Application) 12
- B) Show that canonical LR parser declares error faster than the LALR parser by using following grammar: (Application) 12

$G = (\{E, D\}, \{d, e\}, \{E \rightarrow DD, D \rightarrow dD \mid e\}, E)$

Q.4 Solve Any Two of the following.

- A) Write a short note on YACC. (Knowledge) 06
- B) What is attribute grammar? Write CFG for variable declaration of type integer and float in 'C' language and obtain syntax directed definition from it to assign the declared data type to all variables in declared variable list. Show attribute dependency in parse tree for the string:

int x, y, z

- C) Explain with example the bottom-Up Evaluation of S-attributed definition. (Synthesis) 06

Q. 5 Solve Any two of the following.

- A) Explain the peephole optimization in detail. (Understand) 06
- B) What is basic block and flow graph. Give the algorithm to find flow graph in a three address code. Find flow graph for the following code. (Synthesis) 06

```
read x;
if 0 < x then
    fact := 1;
    repeat
        fact := fact * x;
        x := x-1;
    until x = 0
    write fact;
end
```

- C) Write a note on issues in the design of code generation phase of compiler. (Understand) 06

*** End ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Examination – 2023 Course: B. Tech. Branch : Information Technology Semester : VI Subject Code & Name: BTITC602_Y21 Compiler Construction Max Marks: 60 Date:18/1/2024 Duration: 3 Hr.				
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.				
		(Level/CO)	Marks	
Q. 1	Solve Any Two of the following.		12	
A)	Explain the equivalence between NFA and DFA.	CO1	6	
B)	Design a Turing machine for the palindrome strings.	CO1	6	
C)	Design a pushdown automata for the palindrome strings.	CO1	6	
Q.2	Solve Any Two of the following.		12	
A)	Write a short note on the grouping of phases and also explain its advantages.	CO2	6	
B)	Explain how input buffering is used to recognize the tokens.	CO2	6	
C)	Illustrate the use of RE in lexical analysis with example.	CO2	6	
Q. 3	Solve Any Two of the following.		12	
A)	Explain how ambiguous grammar create difficulties in parser with example.	CO3	6	
B)	Differentiate top-down parser and bottom-up parser.	CO3	6	
C)	Remove left recursion and find FIRST and FOLLOW for the following grammar $G = \{E \rightarrow E+T/T, T \rightarrow T * F/F, F \rightarrow id/(E)\}$	CO3	6	
Q.4	Solve Any Two of the following.		12	
A)	Explain in brief about synthesized and inherited attributes.	CO3	6	
B)	Explain evaluation of S attributed definitions in detail.	CO2	6	
C)	Write a note on importance of syntax tree in compilation.	CO2	6	
Q. 5	Solve Any Two of the following.		12	
A)	Explain the term backpatching with example.	CO3	6	
B)	Explain the term type checking in compilation in detail.	CO2	6	
C)	Write a note on peephole optimization in detail.	CO3	6	
*** End ***				

